

MUHAMMAD SHARIFF BIN ABDUL RASHID

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EDUCATION

Singapore Management University

Singapore

BSc Information Systems (Product Development), 2nd Major in CS (Artificial Intelligence) Aug 2023 – May 2027

- Relevant Coursework: Software Project Management, Enterprise Solution Development, Web App Development, Object-Oriented Programming, Generative AI with LLMs, AI Safety
- Data Associate of the 11th Data Associates Program; SMU Business Analytics & Intelligence Club

EXPERIENCE

HTX (Home Team Science & Technology Agency)

Singapore

Software Engineering Intern @ xDigital (AI Products Team)

Jan 2026 – Present

- Independently building a custom **MCP server** exposing the system's report creation, upload, and lookup functionality as tool contracts for future **agentic AI** integration
- Developing features across a **full-stack TypeScript (React)** codebase for an **AI-powered** procurement report system serving **Whole-of-Government (WoG)**, contributing to report generation workflows
- Implemented **Playwright E2E** test suites in **GitLab CI** workflows to enforce regression coverage across all product features and critical user flows, supporting the product's launch in late January

Zora Health

Singapore

Product Intern

May 2024 – July 2024

- Executed continuous feature delivery for a B2B2C Learning Experience Platform against a strategic roadmap, prioritizing development tasks and validating technical feasibility directly with engineering
- **Automated 90%** of the content production pipeline by developing a **custom Python script**, significantly reducing manual effort and accelerating feature development cycles

PROJECTS

Privacy-Focused Agentic Anonymiser

Oct 2025 – Nov 2025

End-to-end anonymisation pipeline that integrates NER and generative anonymisation into a single workflow

- Built a fully local privacy-preserving multi-agent system with **Ollama** chaining NER detection, generative replacement, and adversarial re-identification to boost anonymisation from **87% to 97%**
- Fine-tuned a **BERT** NER model (Micro-F1 0.87) on OpenPII for PII span detection and implemented Automatic Prompt Engineering with evolutionary mutation to optimise prompts
- Developed a **FastAPI** backend orchestrating **LangChain** pipelines and Transformers inference, with a Next.js frontend supporting DOCX upload and format-preserved download

CampusG Food Delivery Platform

Feb 2025 – Apr 2025

A full-stack, microservices-based food delivery web application connecting customers and delivery runners

- Architected and developed **containerized** microservices (User, Order, Payment, Composites) using **Docker**, each with a dedicated **PostgreSQL** database to ensure service autonomy and scalability
- Engineered data consistency across distributed transactions by implementing the **Saga orchestration** pattern with **Temporal** and **Kafka**, including **compensation** logic for rollbacks
- Integrated **payments** using the **Stripe Connect API** to handle payments directly between customers and runners, using **Clerk** for secure user authentication

ESMOS Migration

Feb 2025 – Apr 2025

An IaaS migration project for a subscription-based meal ordering system to improve scalability and security

- Executed the migration from Odoo SaaS to **Azure Kubernetes Service (AKS)**, designing a scalable IaaS architecture that reduced infrastructure costs by over 40%
- Implemented **Horizontal Pod Autoscaling** for 10,000+ daily users and resolved session-affinity issues with cookie-based sticky sessions, achieving 0% error rate during Locust load testing
- Established a **DevSecOps** pipeline using K8s YAML manifests for **IaC**, Cert Manager for HTTPS, RBAC/NSG for access control, and **Prometheus + Grafana** for observability, ticketing on **Jira**